

Caution

This document is made by AI deepseek from lecture 125 to end on 2 feb 2025
final term

Created by sending reading material available in lectures

For final term pptx ,highlighted reading material and file send msg on
<https://wa.me/+923041069906>

This is only main types short definations to remember

1. Agent-Based Modeling (ABM)

Main Definitions

Agent: An autonomous entity that can perceive its environment, make decisions, and act upon the environment.

Environment: The space or context in which agents operate and interact.

Interaction: The way agents communicate or influence each other and their environment.

Types of Agents

Mobile Agents: Agents that can move around the environment (e.g., turtles in NetLogo).

Stationary Agents: Agents that cannot move (e.g., patches in NetLogo).

Connecting Agents: Agents that link two or more agents (e.g., links in NetLogo).

Key Points to Remember

Agents have properties (e.g., energy, speed) and behaviors (e.g., moving, eating).

ABM is used to simulate complex systems by modeling individual agents and their interactions.

2. NetLogo Basics

Main Definitions

Turtles: Mobile agents in NetLogo.

Patches: Stationary agents that represent the environment.

Links: Connecting agents that represent relationships between turtles.

Key Commands in NetLogo

ask: Used to command agents to perform actions.

let: Used to define variables.

ifelse: Used for conditional statements.

with: Used to filter agents based on conditions.

one-of: Randomly selects one agent from a group.

Key Points to Remember

NetLogo is a popular tool for ABM

The world in NetLogo is divided into patches, and turtles move on these patches.

Links connect turtles to represent relationships (e.g., friendships, communication channels).

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3. Agent Properties and Behaviors

Main Definitions

Properties: Attributes of agents (e.g., energy, speed, color).

Behaviors: Actions agents can perform (e.g., moving, eating, reproducing).

Types of Agent Properties

Global Properties: Properties shared by all agents (e.g., energy in a predator-prey model).

Breed-Specific Properties: Properties unique to a specific breed of agents (e.g., wolves have fang-strength, sheep have wooliness).

Key Points to Remember

Agents can have both shared and unique properties.

Behaviors are defined using rules (e.g., if a wolf has low energy, it eats a sheep).

4. Agent Cognition

Main Definitions

Cognition: The process by which agents make decisions based on their environment and internal state.

Types of Agent Cognition

Reflexive Agents: React to inputs using simple if-then rules (e.g., if there's food, eat it).

Utility-Based Agents: Make decisions to maximize a utility function (e.g., maximize fuel efficiency in cars).

Goal-Based Agents: Have specific goals and take actions to achieve them (e.g., reach a destination).

Adaptive Agents: Learn from past experiences and change their behavior over time.

Key Points to Remember

Reflexive agents are the simplest, while adaptive agents are the most complex.

Utility-based agents optimize a specific function (e.g., fuel efficiency).

Goal-based agents have clear objectives (e.g., reaching home or work).

5. Agent-Environment Interactions

Main Definitions

Agent-Environment Interaction: How agents perceive and modify their environment.

Environment-Self Interaction: How the environment changes over time (e.g., grass regrowth).

Types of Interactions

Agent-Self: Agents interact with themselves (e.g., reproducing).

Agent-Agent: Agents interact with other agents (e.g., wolves eating sheep).

Agent-Environment: Agents interact with the environment (e.g., ants leaving pheromones).

Key Points to Remember

Agents can change the environment (e.g., ants leaving pheromones).

The environment can also influence agent behavior (e.g., pheromones guide ants to food).

6. Complex Adaptive Systems (CAS)

Main Definitions

Complex Adaptive System (CAS): A system where agents interact in ways that lead to emergent behavior (e.g., flocking birds, traffic patterns).

Key Characteristics of CAS

Emergence: Complex patterns arise from simple interactions.

Adaptation: Agents change their behavior based on experience.

Self-Organization: The system organizes itself without central control.

Key Points to Remember

CAS are used to model systems like ecosystems, economies, and social networks.

Emergent behavior is a key feature of CAS (e.g., traffic jams emerge from individual car behaviors).

7. Validation and Verification in ABM

Main Definitions

Validation: Ensuring the model accurately represents the real-world system.

Verification: Ensuring the model is implemented correctly (e.g., no bugs in the code).

Key Techniques

Sensitivity Analysis: Testing how changes in parameters affect the model.

Calibration: Adjusting model parameters to match real-world data.

Replication: Running the model multiple times to ensure consistent results.

Key Points to Remember

Validation ensures the model is realistic.

Verification ensures the model works as intended.

Sensitivity analysis helps identify critical parameters.

8. Applications of Agent-Based Modeling

Main Applications

Disease Spread: Modeling how diseases spread through a population.

Traffic Simulation: Simulating traffic flow and congestion.

Predator-Prey Dynamics: Modeling interactions between predators and prey.

Social Networks: Simulating how information spreads through a network.

Key Points to Remember

ABM is used in diverse fields like biology, sociology, and urban planning.

Real-world applications include predicting disease outbreaks and optimizing traffic systems.

9. Key Concepts to Remember

Agents: Autonomous entities with properties and behaviors.

Environment: The space where agents operate.

Interactions: How agents communicate and influence each other.

Cognition: How agents make decisions (reflexive, utility-based, goal-based, adaptive).

Emergence: Complex patterns arising from simple interactions.

Validation and Verification: Ensuring the model is accurate and correctly implemented.

NetLogo: A tool for creating and simulating ABMs.
